# Other Assumptions about Errors

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# Other assumptions about errors

- In our preceding discussions of errors and distributions, we made several implicit assumptions.

- We need to make those assumptions explicit so that we know (1) precisely what those assumptions are and (2) situations in which those assumptions might be violated.

# 1. Independence

- Independence is the assumption that any given error term is not systematically related to another error term.
  - I.e., independence is violated when knowing one error term allows us to make a reasonable guess about another error term.
  - An example of this would be ***time-series*** data. A player's performance in 2017 is more probably more predictive of their performance in 2018 than their 2009 performance.
  - Another example would be ***clustered*** data. Individuals from the same family are more like each other (usually) than individuals from different families.
    - More broadly, these are both examples of ***nested*** data. In the time-series case there is a natural order to the data, whereas in the clustered case, there is not.

# 1. Independence

- Our assumption of normality is relatively ***robust***. That is, as long as the underlying distribution is approximately normal, we will get a similar answer as if we had a truly normal distribution.

- ***The independence assumption is not at all robust***. That is, if we fail to recognize statistical dependence in our data we will get a very different (i.e. very wrong) answer. Thus, we need to properly identify and handle statistical dependence.
  - We will discuss how to address statistical dependence in later chapters.

# 2. Identically Distributed

- We assume that all errors are sampled from the same distribution.
- For instance, if we randomly sampled two groups from the population, we would assume that these groups have similar (or ***homogenous***) variances.
- Or, if I subjected people to different treatments (i.e., placebo versus a caffeine supplement) their average level of performance might change, but variability in their performance might increase as well.
- If the independence and identical distribution assumptions are met, then the errors are referred to as ***IID*** (independent and identically distributed).

# 3. Errors are Unbiased

- Finally, we make an assumption that the distribution of errors is unbiased (i.e., the mean of all errors is 0).

- This assumption is important because, if the distribution of errors averaged either above or below zero, then some systematic error is being forced into the model (i.e., DATA = MODEL + ERROR).

- Depending on the substantive domain, ensuring that errors are random involves aspects of the experimental design, measurement tools, and 'missingness'.
  - We will generally assume that data are missing at random (MAR). In practice this is not always the case and data might be missing not at random (MNAR).
  - I.e., in a longitudinal study, data are more likely to be missing from the end of the study than the beginning.